



Supply Chain Security, Procurement Continuity

Presenting to: BC Hydro

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**Objective**

Identify the category most vulnerable to supply chain disruption and consider strategic approaches for the short and long term

Problems

Immediate procurement
vulnerability across all
categories

Category vulnerability from
USA imports

Approach a 25% USA tariff in
both **short** and **long term**

Strategy**Smart Sourcing****Resilience**

Target low buffer, low-risk
categories

Value

Diversify portfolio for future
growth and security

**Key
Takeaway**

Leveraging global vendors in the short term will protect vulnerable categories, while long term partnerships in Europe will expedite procurement delivery



Data analysis break down for efforts focused on highly important, highly vulnerable categories

Methodology

Low Risk Tolerance



Prioritizing low risk tolerance categories as an operational necessity

Buffer Stock



Buffer stock in days as a **proxy for vulnerability**
(=Total days of supply – Avg. Lead time)

Lead Time



Lead time as the main decision criterion for supply chain analysis **in highly vulnerable categories**

Performance
& Tariffs



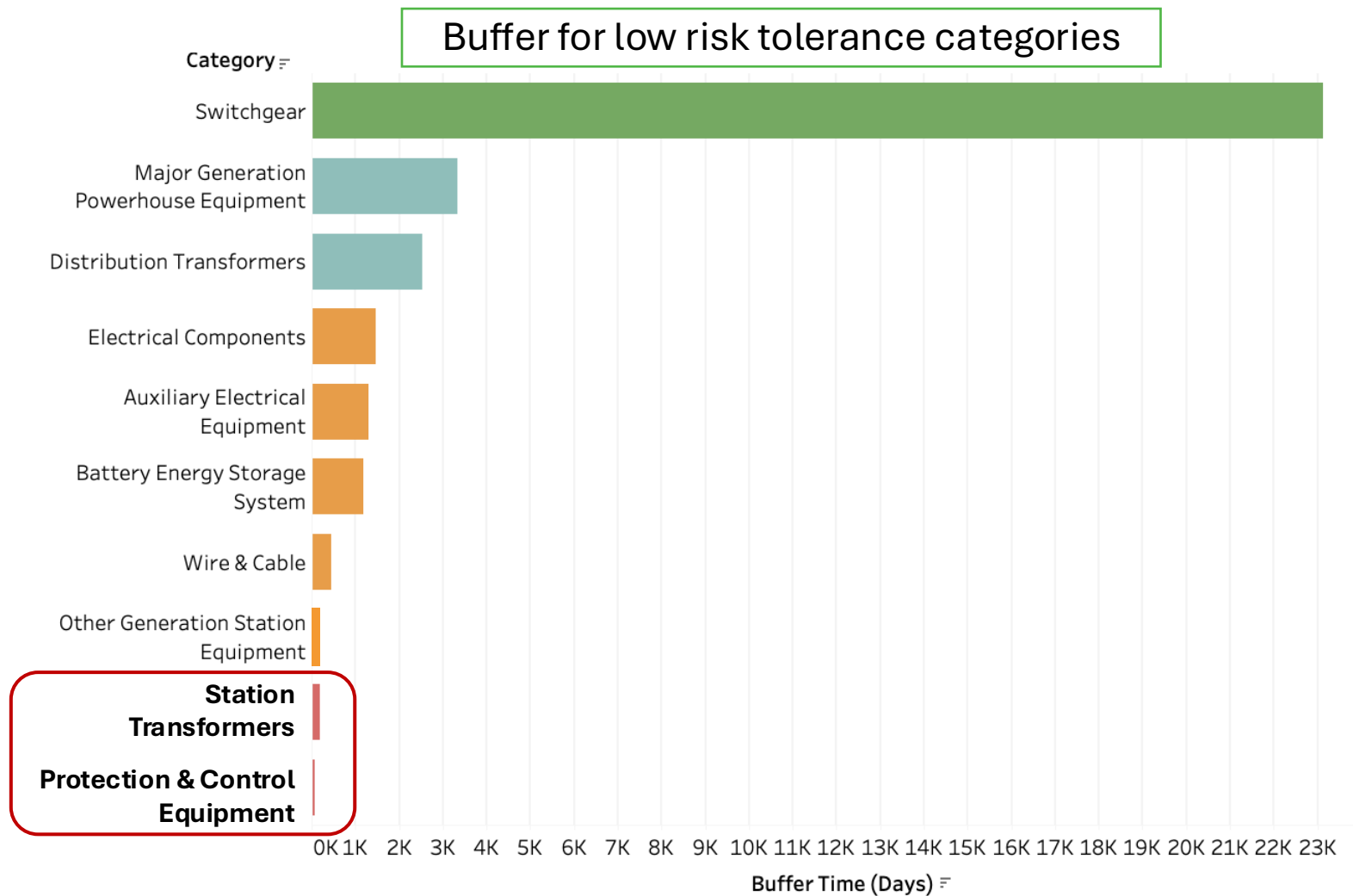
Factoring **vendor performance (1-5)** and **tariff rates**



Buffer stock measure vendors active inventory, frequency of use, and average lead time

Buffer stock accounts for

- Vendors active inventory
- Frequency of use
- Average lead time





Two low risk tolerance categories show vulnerability from low buffer and long lead times

Buffer stock accounts for

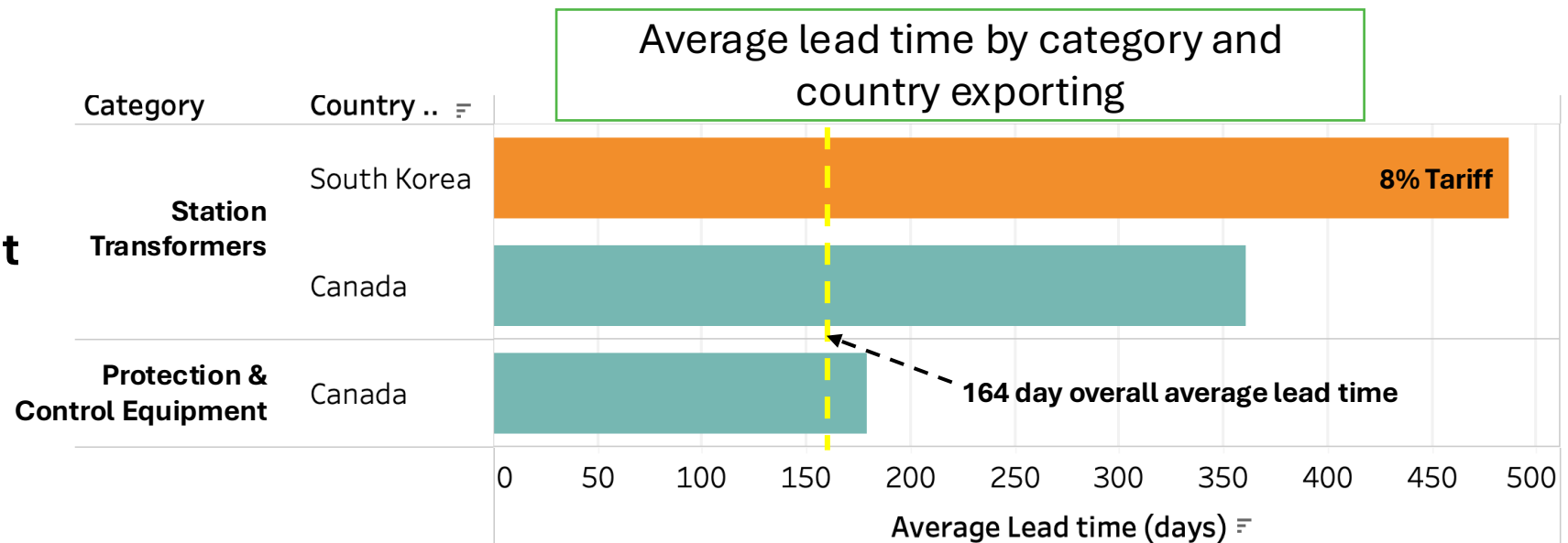
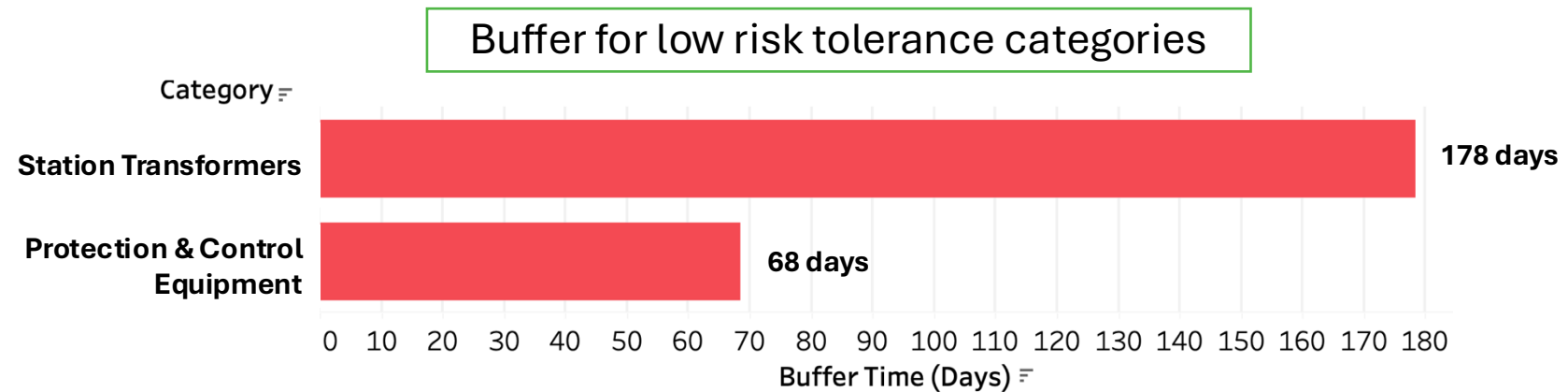
- Vendors active inventory
- Frequency of use
- Average lead time

Station Transformers

- Long Lead Times
- Subject to 8% Korean Tariff

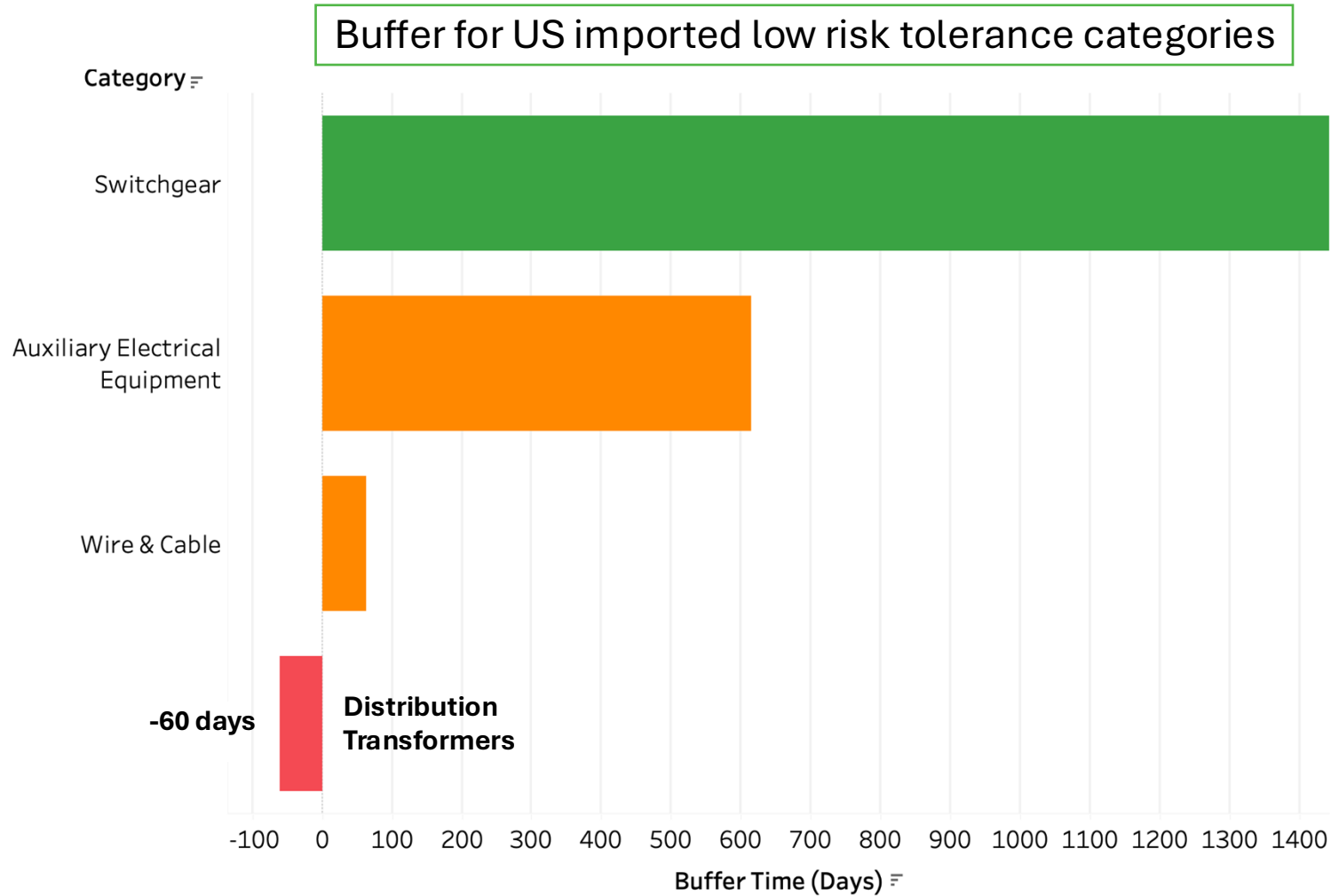
Protection & Control Equipment

- Lowest buffer stock
- Reliant on **sole vendor**
- Low safety stock (12)





Distribution transformer inventory will be empty for 2 months without a vendor change

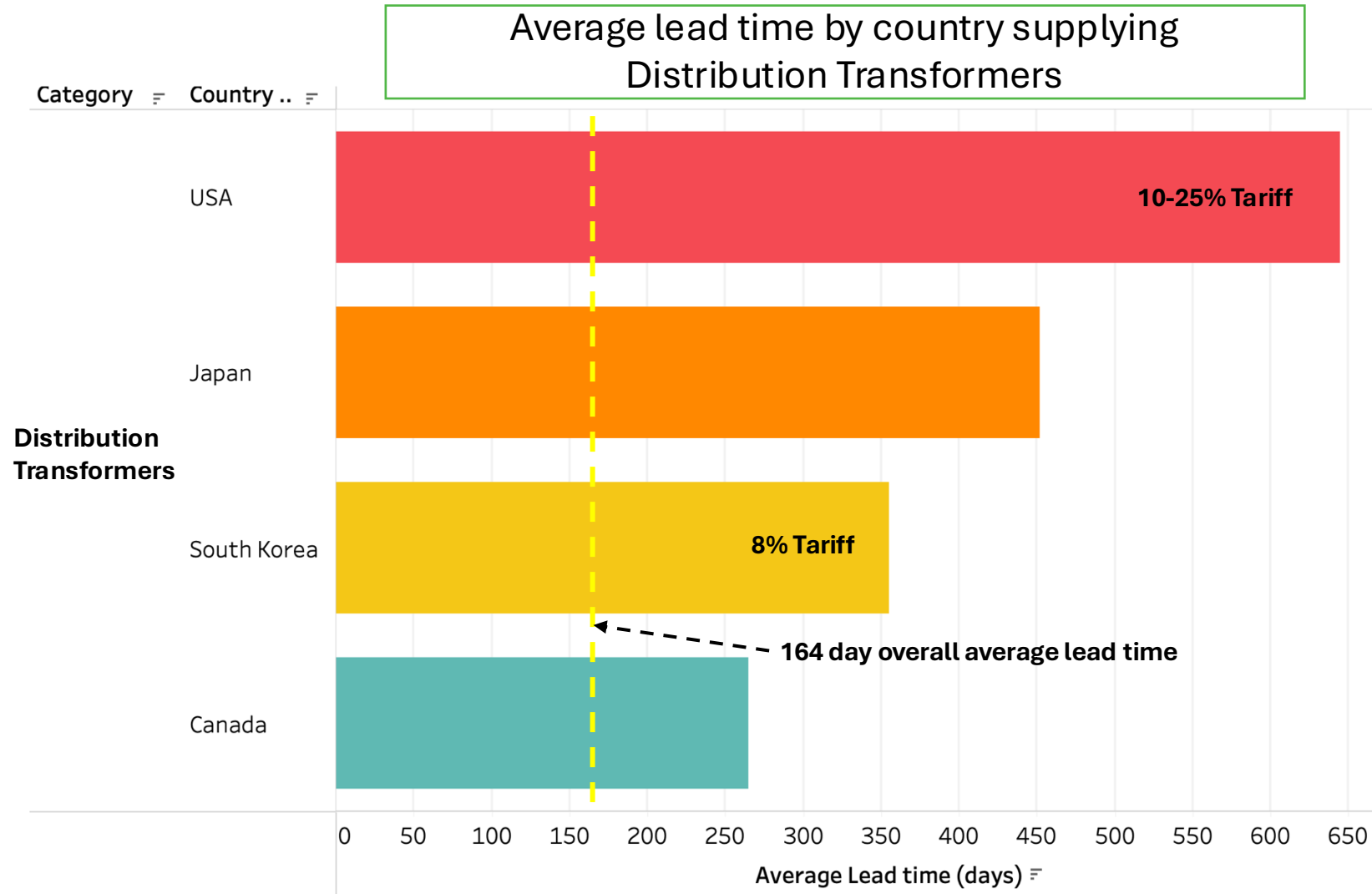


Distribution Transformers

- -60 day buffer means **no stock** for 2 months
- Large standard deviation +/- 40% of average lead time



BC Hydro has vendors options around the world for Distribution Transformers



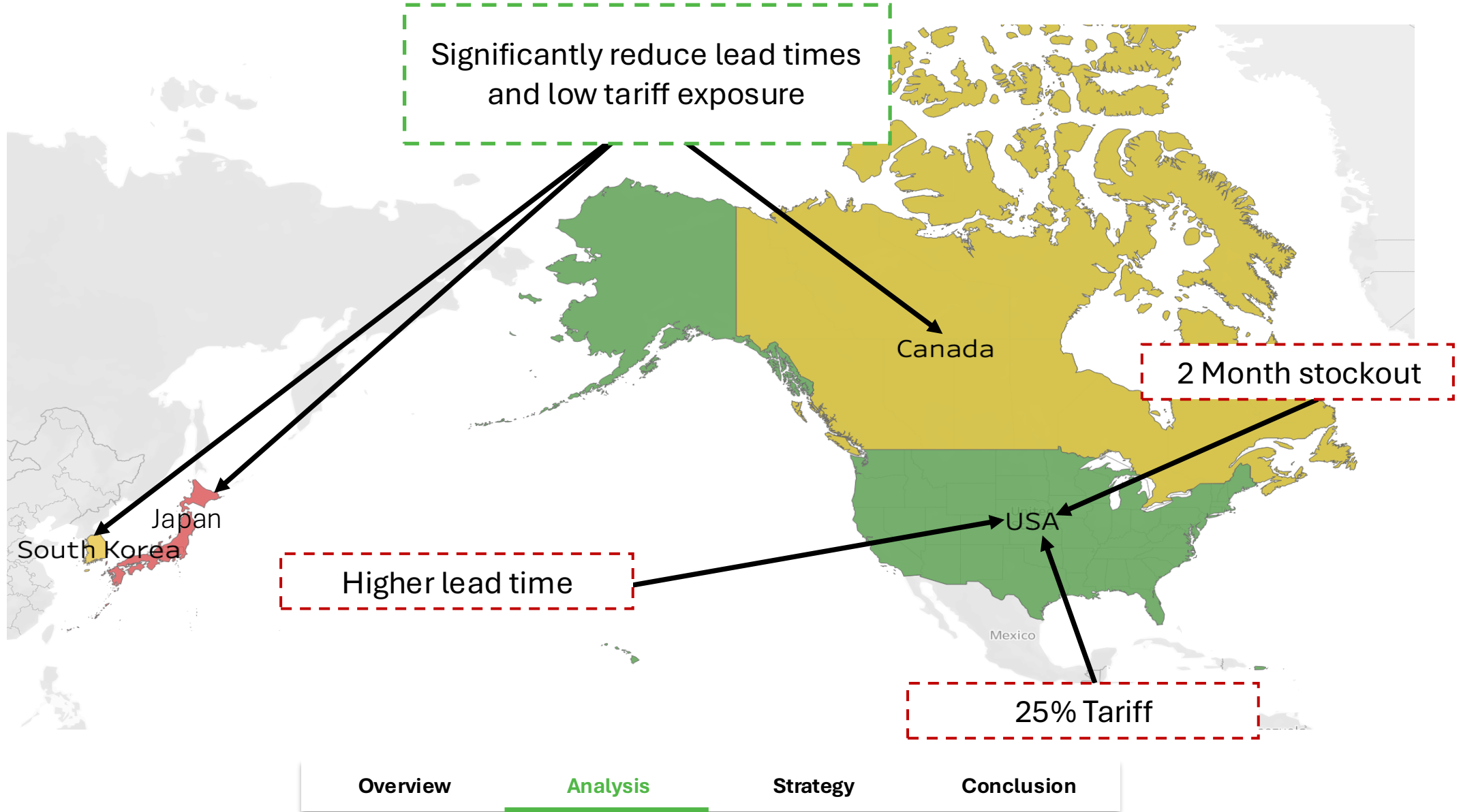
Distribution Transformers

- -60 day buffer means **no stock** for 2 months
- Large standard deviation +/- 40% of average lead time

Strategic Opportunity

- Japan, South Korea, and Canada offer **lower tariff rates and lead times**

Transition distribution transformer procurement to South Korea and Canada to minimize lead times and tariff exposure

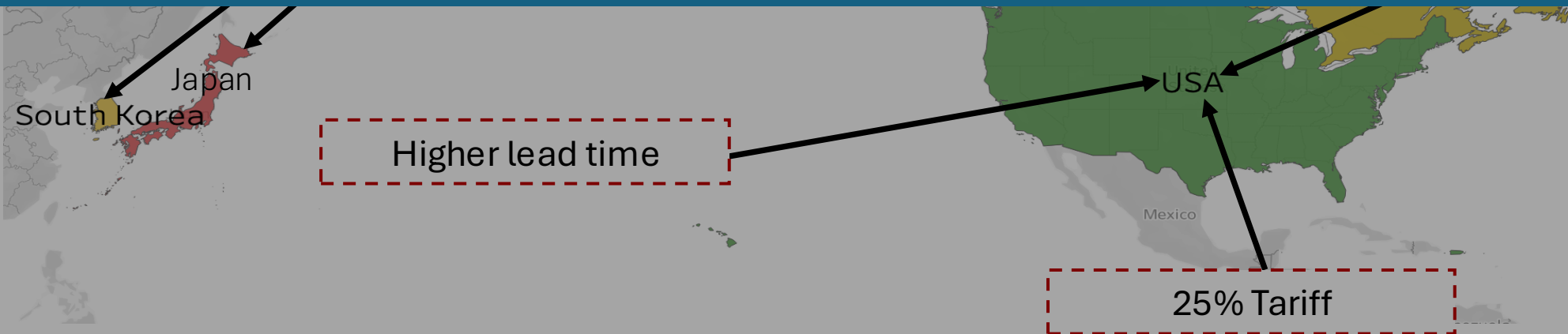


Transition distribution transformer procurement to South Korea and Canada to minimize lead times and tariff exposure

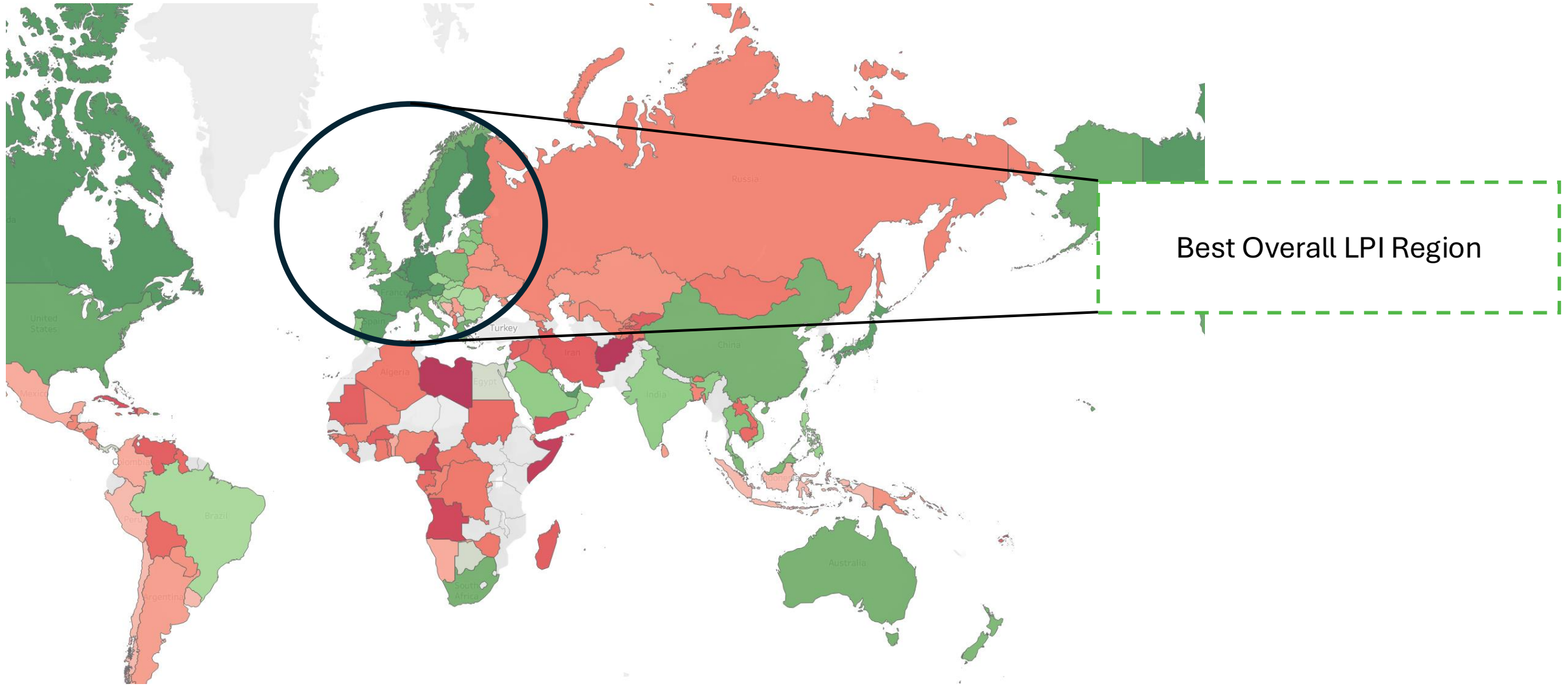


Significantly reduce lead times
and low tariff exposure

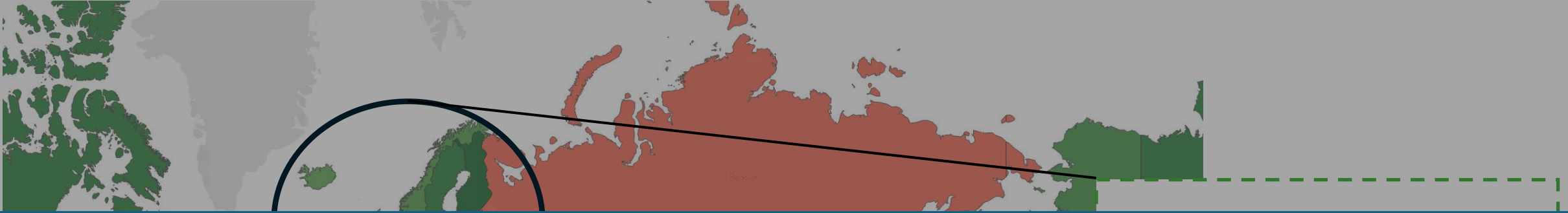
Prioritize sourcing distribution transformers from **South Korea and Canada** instead of the U.S. to significantly reduce lead times and tariff exposure



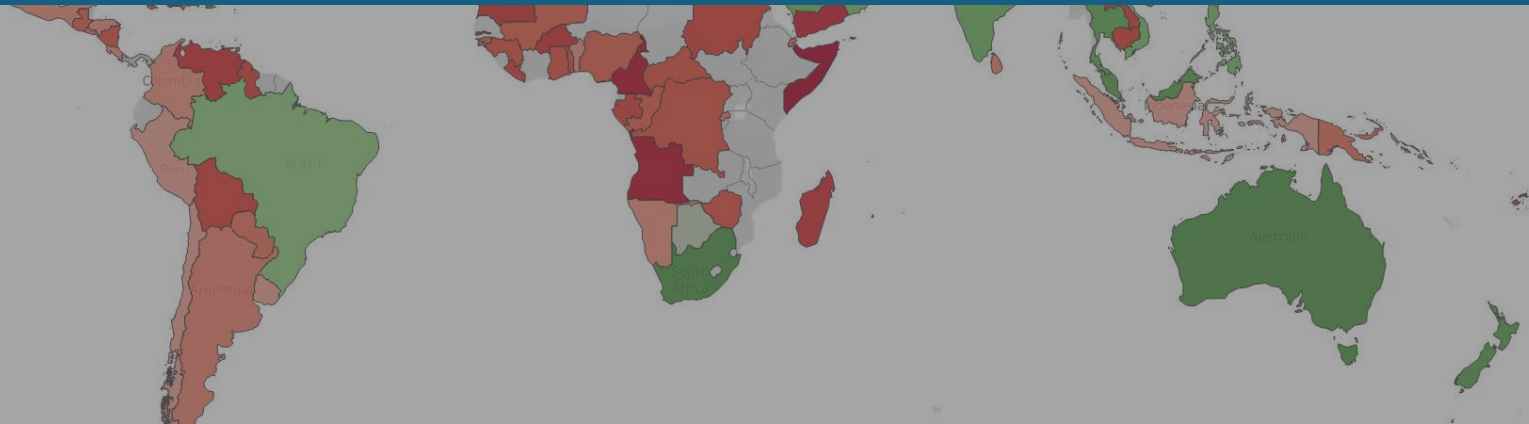
Considering potential alternatives based on LPI



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Urgent need to diversify into Europe with potential increasing USA tariff



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Appendix

Short-term Strategy

**KEY ISSUE:**

- **Distribution transformers** have a buffer time of ~60 days, worst case there could be no on-hand stock for **two months**
- **Vendor Pool:** Available suppliers are in the USA, Japan, South Korea, and Canada
- Although USA vendors have the best performance, **longer lead times and 25% tariffs** is why we need alternatives

High lead time + High tariffs + Negative buffer = Significant stockout and Cost risk

**Result:**

Two-Month Stockout Risk
Protracted Replenishment
Escalated Procurement Costs



Prioritize sourcing distribution transformers from South Korea and Canada instead of the U.S. to significantly reduce lead times and tariff exposure

